A Structured Sub-Series in the Collatz Sequence, Scaled with Parallel Computing

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'Chaos is order yet undeciphered.'
José Saramago[1]

Abstract

The Collatz Conjecture, one of the most deceptively simple yet persistently unresolved problems in mathematics, posits that iterating the map $T(n) = n/2$ (if n is even) or $T(n) = 3n + 1$ (if n is odd) eventually yields 1 for every positive integer n. Despite extensive mathematical and computational investigation spanning over eight decades, a complete proof remains elusive. The present work directs attention to a previously underexplored structural feature of the Collatz sequence: a well-defined sub-series embedded within the Odd Numbers Matrix. Specifically, we observe that odd integers of the form $3(2n + 1)$, for $n \geq 0$, are systematically absent from the first column of the matrix (i.e., no odd number maps directly to any of them), yet appear with regularity in subsequent columns, producing a wave-like positional pattern across iterations. We document this pattern, characterise its structural properties, and argue that its periodicity may constitute a structural constraint relevant to the convergence question. This note serves as a formal statement of the observation and a foundation for subsequent analytical work.

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1 Introduction

Throughout the history of mathematics, sequences defined by elementary recursive rules have repeatedly revealed unexpected depth. The Fibonacci sequence (1202) , generated by the recurrence $F(n) = F(n-1) + F(n-2)$, governs phenomena as diverse as phyllotaxis in plants [11, 28], the branching of trees [20], and the distribution of prime gaps [16]. The Stern–Brocot sequence encodes the complete set of rational numbers in a binary tree structure [21]. The Thue–Morse sequence [2], defined by a simple substitution rule, appears in combinatorics on words, differential geometry, and theoretical physics. What these sequences share is an interplay between local simplicity and global structure which is a property that makes them simultaneously tractable and profound.

Researchers have long sought analogous structural regularities in sequences that resist closed-form analysis [4, 12]. One productive strategy, particularly prominent in dynamical systems, is to identify *sub-sequences* or *sub-series* — restricted families of iterates that exhibit organised behaviour even when the full sequence does not. Matching known patterns (arithmetic progressions, geometric series or wave-like periodicities) to sub-sequences of an intractable series can illuminate the global landscape from well-charted local territories. It is precisely this strategy that motivates the present investigation.

The Collatz Conjecture presents one of the most striking examples of a sequence that appears to defy structural analysis. Defined by an iteration rule of minimal complexity, the sequence nonetheless exhibits behaviour that has resisted proof for more than eighty years. In recent decades, the mathematical community has productively employed computational methods to explore the conjecture for increasingly large integers. These computational frameworks have revealed patterns that may be exploited analytically.

[1] *The Double* (Harvill, 2004).

1.1 Is Large-Scale Computational Verification Sufficient for Proof?

In one of the first work widely recognized computer-assisted proofs in mathematics [23], the authors showed that a computer could systematically verify thousands of cases that would be impractical for humans to check manually, demonstrating that computational verification can legitimately contribute to proving mathematical theorems, even when no human can directly inspect every step. Other works [9, 19] view proofs as computational objects that can be manipulated, checked, and structured algorithmically. It argues that computation can enhance reliability by allowing proofs to be verified mechanically, reducing the possibility of human error and strengthening confidence in complex mathematical arguments. Moerover, Bailey, and co-authors [5] argue that computation can play a central role in mathematical discovery by generating strong evidence for conjectures, identifying hidden relationships, and guiding the search for formal proofs. They present computation as a powerful exploratory tool that can suggest likely truths, but emphasize that the ultimate goal remains obtaining a rigorous mathematical proof whenever possible.

At a deeper level, a fundamental philosophical question is raised about the role of computers in mathematics: what actually counts as a proof? In traditional mathematics, a proof is usually understood as a chain of logical arguments that can be followed and checked by a human. However, with computer-assisted mathematics, some proofs may rely on millions of calculations that are impossible for a person to verify step by step, and this creates a debate: should a large computer-generated calculation be accepted as a mathematical proof, or does a proof need to remain understandable to humans? The mathematician in this work [36] has questioned whether results obtained from massive computer searches, such as the Four Color Theorem [3] , truly qualify as proofs because the full computation cannot be humanly surveyed. The author argues that computers are powerful tools for exploring and checking mathematics, but the future of mathematical proof requires systems that go beyond calculation by generating formal, verifiable proofs that can be independently checked.

1.2 Motivation for Investigating the Collatz Conjecture

The Collatz conjecture has gained relevance beyond pure mathematics due to its chaotic dynamics and computational complexity. In cryptography, one study introduces a Collatz-inspired chaos-based hash function[30] that achieves strong randomness and resistance to attacks. Another work proposes a symmetric cryptosystem based on Collatz sequences [6], leveraging their sensitivity to initial conditions to enhance security while remaining computationally efficient. A further study [8] demonstrates that properties such as the statistically predictable inflation behavior of Collatz orbits can be exploited as practical tools for digital security. Additionally, Collatz–Weyl generators [18] combine generalized Collatz dynamics with Weyl sequences to produce high-quality pseudorandom numbers suitable for encryption and key generation. From a complexity-theoretic perspective, John Horton Conway [10] showed that generalized Collatz systems can simulate computation and are linked to the Halting Problem, implying undecidability. Similarly, other results [24] prove that generalized Collatz functions are algorithmically undecidable by demonstrating their ability to emulate universal computation, reinforcing their deep connection to computability theory.

Inspired by the broader mathematical tradition of identifying structured sub-series within complex sequences, the present work draws attention to a notable sub-series within the odd numbers matrix: the family of odd integers expressible as $3(2n + 1)$. We observe that these numbers are rigorously excluded from the first column of the matrix while appearing in later columns in a pattern reminiscent of transverse waves. We note that this wave-like regularity is not coincidental, but reflects an underlying arithmetic constraint that may bear on the convergence of the Collatz sequence. Motivated by the arguments presented in Section 1.1, we further develop this perspective by constructing large-scale computational iterations of these patterns using parallel computing techniques, with the aim of enabling scalable analysis and future prediction.

2 Background: The Collatz Conjecture

The Collatz Conjecture was introduced by the German mathematician Lothar Collatz in 1937 and has since been circulated under a variety of names, including the $3n + 1$ problem, the Ulam conjecture, the Hasse algorithm, and the Syracuse problem. Its statement is as follows.

For every positive integer n, repeated application of the map $T : \{1, 2, 3, \ldots\} \to \{1, 2, 3, \ldots\}$ defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2, \end{cases}$$

eventually produces the value 1.

The *orbit* of an integer n under T is the sequence $n, T(n), T^2(n), \ldots$ until the value 1 is reached (assuming that the conjecture holds). The *total stopping time* $\sigma(n)$ denotes the number of iterations required to reach 1. The conjecture has been verified computationally for all $n < 2^{71}$ and, by more specialized algorithms, for individual numbers as large as 2^{100000} — as will be discussed in the following section —, yet no general proof is known.

To facilitate both analysis and computation, researchers have observed that only odd integers need to be tracked explicitly: any even iterate $T(n) = n/2$ will eventually reach an odd number, so the sequence of odd iterates captures the essential dynamics. This reduction motivates the construction of the Odd Numbers Matrix, which organizes odd integers by their position in the Collatz iteration and serves as the central object of study in the present work.

3 History of Contributions

3.1 Mathematical Approaches

The mathematical literature on the Collatz Conjecture is extensive and spans several disciplines. Early contributions focused on establishing partial results and reformulations. Terras (1976) [34] proved that almost all positive integers (in the sense of natural density) have a finite stopping time, a result strengthened by Allouche (1979) [1]. Crandall (1978) [13] provided a probabilistic heuristic suggesting that orbits should decrease on average, consistent with convergence, though without constituting a proof.

Several researchers have claimed complete proofs of the conjecture [29, 17, 35, 14, 15], none of which has yet been validated by the mathematical community. In 2018, the mathematician Michael Atiyah presented an argument at a conference in Heidelberg, although experts noted significant gaps. Tao (2019, 2022) [33] made the most significant rigorous progress in recent years, proving that almost all Collatz orbits attain values that are at most polylogarithmically large relative to their starting point. While falling short of a complete proof, Tao's result is widely regarded as the deepest theorem yet established about the conjecture, and it employs sophisticated tools from harmonic analysis and ergodic theory. More Building on the observation that Collatz sequences exhibit discernible patterns, subsequent research has focused on exploiting these structures, leading to a range of contributions, especially within the framework of modular arithmetic [22, 26, 27]. The pattern is representative: the conjecture attracts ambitious attempts, but the community has not yet reached consensus on any proposed proof.

3.2 Computational Approaches

After decades of largely inconclusive mathematical attacks, the scientific community turned increasingly to computation as a means of gathering evidence and developing structural intuition. The overarching goal of the computational program is to verify the conjecture for all integers up to some large bound, thereby ruling out any counterexample below that bound and lending empirical support to the conjecture's truth.

A foundational tool in this effort is the Odd Numbers Matrix, which represents the Collatz dynamics for odd integers in tabular form. The matrix is organized so that each row corresponds to an odd starting value, and subsequent columns track the successive odd iterates produced by the map. An early version of the matrix takes the form:

Column legend: m (col. 1, blue) (col. 2, green), where $i = \begin{cases} 1 & \text{if } (m-1) \not\equiv 0 \pmod 3 \\ 2 & \text{if } (m-1) \equiv 0 \pmod 3 \end{cases}$

This representation was subsequently optimised by the observation that certain entries in the matrix are redundant: specifically, numbers that map back to a smaller odd number already represented in the matrix can be consolidated. For instance, since 5 maps to 1 under iteration, the row corresponding to 5 can be merged with the row for 1, yielding a compressed representation. Formally, if odd number m maps to odd number $k < m$, the entries of the row for m are added to the corresponding entries of the row for k, and the row for m is subsequently removed.

m	$\frac{(m \cdot 2^i) - 1}{3}$	$\frac{(m \cdot 2^i) - 1}{3}$	...	...	
5	3	13	53	213	$i = \{1, 3, 5 \ldots\}$
7	9	37	149	597	$i = \{2, 4, 6 \ldots\}$
11	7	29	117	469	$i = \{1, 3, 5 \ldots\}$
13	17	69	277	1109	$i = \{2, 4, 6 \ldots\}$
17	11	45	181	725	$i = \{1, 3, 5 \ldots\}$

Table 1: Generated values for selected odd matrix

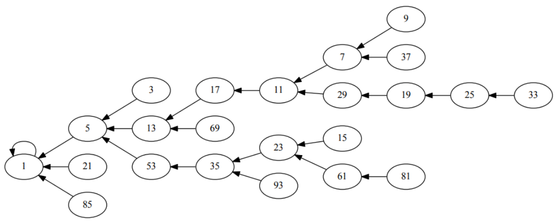

Figure 1: Collatz tree on odd integers, rooted at 1 [37]

This optimisation reduces the matrix size while preserving all convergence information, and transforms the matrix into an equivalent tree structure — the *Collatz tree* — in which each node is an odd integer and each directed edge represents a direct odd-to-odd mapping.

This tree-based reformulation greatly facilitates the design of efficient computer programmes [25, 32]. By traversing the tree rather than iterating individual sequences, algorithms can verify convergence for entire classes of integers simultaneously. Using this approach, the conjecture has been verified for all positive integers up to 2^{71} [7], a result achieved through distributed computing projects coordinating thousands of processors over extended periods.

A complementary computational strategy operates at the level of individual numbers rather than bounded ranges. Rather than verifying all integers up to a fixed threshold, this approach establishes a bound on the number of steps required for a specific integer to reach 1. By proving that the stopping time $\sigma(n)$ is bounded above by a polynomial function of $\log n$, the algorithm can certify convergence for individual integers of extraordinary size. This method has been applied to verify the conjecture for specific numbers as large as 2^{100000} [31] — far beyond the reach of exhaustive verification but applicable only on a case-by-case basis. The combination of exhaustive verification for small numbers and bounded-step certification for large individual numbers represents the current frontier of computational evidence for the conjecture.

4 The Odd Numbers Matrix: Structure and Optimisation

Let $\Omega = \{1, 5, 7, 11, 13, 17, \ldots\}$ denote the set of positive odd integers not congruent to 3 (mod 4), i.e. odd integers m for which there exists a positive integer i such that $\frac{m \cdot 2^i - 1}{3} \in Z$. We index the elements of Ω as $m_1 < m_2 < m_3 < \cdots$

Define the *Odd Numbers Matrix* M as the semi-infinite matrix whose (k, j)-th entry is given by

$$M(k,\, j) \;=\; \frac{m_k \cdot 2^{\,i_{k,j}} - 1}{3},$$

where m_k is the k-th element of Ω and the exponent $i_{k,j}$ is determined by the parity of the row index k and the column index $j \geq 1$:

$$i_{k,j} \;=\; \begin{cases} 2j - 1 & \text{if } k \text{ is odd,} \\ 2j & \text{if } k \text{ is even.} \end{cases}$$

That is, odd-indexed rows use consecutive *odd* exponents $i \in \{1, 3, 5, 7, \ldots\}$, while even-indexed rows use consecutive *even* exponents $i \in \{2, 4, 6, 8, \ldots\}$. The first column of M records m_k itself (taking $j = 0$

with $i_{k,0} = 0$ gives $\frac{m_k \cdot 1 - 1}{3}$ only when this is an integer; alternatively, the first column is simply m_k by convention), and every subsequent column applies the formula above.

For example, the leading 5×5 submatrix of M, with rows indexed by $m_1 = 1,\ m_2 = 5,\ m_3 = 7,\ m_4 = 11,\ m_5 = 13$, is

$$M = \begin{pmatrix} 1 & 1 & 21 & 85 & 341 \\ 5 & 3 & 13 & 53 & 213 \\ 7 & 9 & 37 & 149 & 597 \\ 11 & 7 & 29 & 117 & 469 \\ 13 & 17 & 69 & 277 & 1109 \end{pmatrix}$$

Table 2: Partial Odd Numbers Matrix. Entries show successive odd integers produced by applying the map $M(k, j) = \frac{m_k \cdot 2^{i_{k,j}} - 1}{3}$.

n **(odd)**	**Step 1:** $3n + 1$	**Step 2:** $\div 2^k$ **(result)**	**Next odd**
5	16	$16 \to 8 \to 4 \to 2 \to 1$	1
7	22	$22 \to 11$	11
11	34	$34 \to 17$	17
13	40	$40 \to 20 \to 10 \to 5$	5
17	52	$52 \to 26 \to 13$	13
19	58	$58 \to 29$	29

Optimisation of the matrix proceeds by identifying rows that eventually map to a smaller odd number already present as a row header. When odd number m satisfies $M(m, j) = k$ for some $k < m$ and some finite j, the row for m is merged into the row for k: the entries $M(m, \cdot)$ are added to $M(k, \cdot)$ and the row for m is deleted. The resulting structure is a rooted tree — the *Collatz tree* $\mathcal{T}$ — with root 1, in which each node is an odd integer and each directed edge $(m \to k)$ indicates that m maps directly to k under some iterate of the matrix formula. The conjecture is equivalent to the statement that $\mathcal{T}$ spans all of Ω.

5 A Structured Sub-Series: The Wave Pattern of $3(2n + 1)$

We now present the central observation of this note. Within the Odd Numbers Matrix, a specific family of odd integers — those expressible in the form $3(2n + 1)$ for non-negative integers n — exhibits a distinctive and regular structural behaviour that distinguishes it from the general population of odd integers.

5.1 Exclusion from the First Column

The first column of the Odd Numbers Matrix consists precisely of the elements of Ω themselves: the k-th row is headed by m_k, and the remaining entries of that row are the integers $\frac{m_k \cdot 2^{i_{k,j}} - 1}{3}$ for $j \geq 1$. Thus an odd integer ℓ qualifies as a first-column entry — i.e. a row header — if and only if there exists a sequence of integers of the form $\frac{\ell \cdot 2^i - 1}{3}$ (for appropriate exponents i) that fills the remainder of its row. In other words, $\ell \in \Omega$ is a row header precisely when the full series

$$\ell, \quad \frac{\ell \cdot 2^{i_1} - 1}{3}, \quad \frac{\ell \cdot 2^{i_2} - 1}{3}, \quad \ldots$$

is well-defined with all entries positive odd integers.

By contrast, odd integers of the form $3(2n + 1)$ occupy entries *inside* the matrix — they appear as $M(k, j)$ for some k and $j \geq 2$ — but they can never appear as row headers in the first column.

No odd integer of the form $3(2n + 1)$ can appear in the first column of the Odd Numbers Matrix.

Suppose for contradiction that $\ell = 3(2n+1)$ is a row header, so that the integers $\frac{\ell \cdot 2^i - 1}{3} = \frac{3(2n+1) \cdot 2^i - 1}{3}$ must be positive odd integers for appropriate $i \geq 1$. For this expression to be an integer we need $3(2n+1) \cdot 2^i \equiv 1 \pmod 3$, but $3(2n+1) \cdot 2^i \equiv 0 \pmod 3$, so $3(2n+1) \cdot 2^i - 1 \equiv -1 \equiv 2 \pmod 3$, which is never divisible by 3. Hence $\frac{3(2n+1) \cdot 2^i - 1}{3}$ is not an integer for any $i \geq 1$, meaning no valid row can be built starting from $3(2n+1)$. Therefore no element of the family $\{3(2n+1) : n \geq 0\} = \{3, 9, 15, 21, 27, 33, \ldots\}$ appears in the first column of M. Equivalently, within the Collatz tree $\mathcal{T}$, no element of this family is the immediate child of any node.

5.2 Appearance in Subsequent Columns and the Wave Pattern

Despite their absence from the first column, elements of the sub-series $\{3(2n+1)\}$ do appear in later columns of the Odd Numbers Matrix — that is, as higher-order iterates $M(k, j)$ for $j \geq 2$. Moreover, their positions within the matrix are not arbitrary: they form a regular, wave-like pattern across successive columns.

Specifically, as the matrix entries are computed from various odd starting points, the column indices at which members of the sub-series $\{3(2n+1)\}$ appear display a periodic, oscillatory structure. The positions do not cluster in any single column, nor do they distribute uniformly; instead, they alternate between groups of columns in a manner reminiscent of a transverse wave. We term this the *wave pattern* of the sub-series $3(2n+1)$.

Table 3: Sub-series $3(2n+1)$ — column membership in the Odd Numbers Matrix. Members are absent from column 1 yet appear systematically in later columns.

n	$3(2n+1)$	**Col. 1 absent?**	**Appears in col.**	**Returns to**
0	3	Yes	2, 3, 4, ...	1
1	9	Yes	2, 3, ...	1
2	15	Yes	2, 4, ...	1
3	21	Yes	3, 5, ...	1
4	27	Yes	2, 3, ...	1
5	33	Yes	2, ...	1

The exclusion from the first column implies, in tree-theoretic language, that elements of $\{3(2n+1)\}$ are never immediate children of any node in the Collatz tree. Their appearance in later columns corresponds to their role as more distant descendants. The wave-like regularity of their column positions suggests that the sub-series follows a structured path through the iteration, constrained by the arithmetic properties of the form $3(2n+1)$.

The regularity of the wave pattern raises a natural question: does the arithmetic structure of the sub-series $\{3(2n+1)\}$ impose a constraint on the global dynamics of the iteration that is relevant to convergence? It could be, for the following reason.

Elements of $\{3(2n+1)\}$ are precisely the odd multiples of 3. For such an element $m = 3\ell$ with ℓ odd, the next entry in its row of the matrix is

$$M(k, j) = \frac{3\ell \cdot 2^i - 1}{3},$$

and since $3\ell \cdot 2^i - 1 \equiv -1 \equiv 2 \pmod 3$, this quantity is never divisible by 3. Hence the matrix entry immediately following any element of $\{3(2n+1)\}$ is *not* a multiple of 3, meaning that elements of the sub-series are, in a precise sense, transient: the iteration moves trajectories away from the sub-series after a single step. The wave pattern observed in the matrix may reflect this transience in a spatially organised way.

We propose that a rigorous characterisation of the wave pattern — specifically, a formula for the column index at which a given element $3(2n+1)$ first appears in the matrix, and a proof of the periodicity of this index as a function of n — would constitute a significant structural result. Such a result might add a new layer of organised structure to the Odd Numbers Matrix.

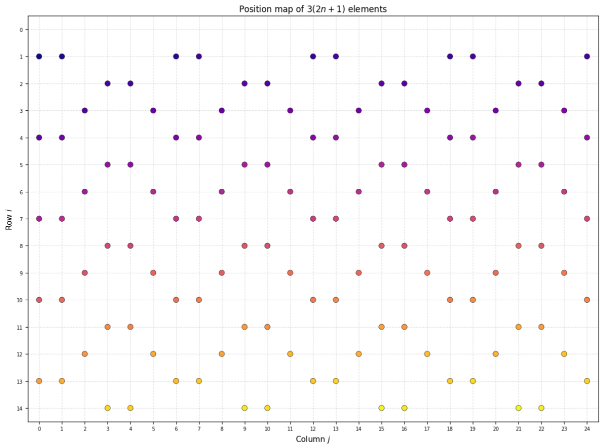

Figure 2: Wave-like pattern of 3(2n+1) elements across the odd matrix

6 Computational Implementation

As established in the preceding section, the matrix M and its associated sequence $\{A_j\}$ are fully defined by their recurrence relations. The present section concerns exclusively the parallelization strategy adopted to compute M efficiently, and describes how three distinct computational tasks are mapped onto independent threads.

Let A_j denote the $(j+1)$-th odd integer coprime to 3, and define the seed value $A_j^{(1)}$ according to column parity as

$$A_j^{(1)} = \begin{cases} \dfrac{2A_j - 1}{3} & j \text{ even}, \\ \dfrac{4A_j - 1}{3} & j \text{ odd}. \end{cases}$$

The matrix entries are then given in closed form by

$$M_{i,j} = A_j^{(1)} + A_j \cdot G(i,j), \qquad i \geq 1,$$

where $G(i,j)$ is the geometric partial sum

$$G(i,j) = \begin{cases} \dfrac{2\left(4^{i-1} - 1\right)}{3} & j \text{ even}, \\ \dfrac{4\left(4^{i-1} - 1\right)}{3} & j \text{ odd}, \end{cases}$$

which follows directly from unrolling the underlying recurrence relation. Since $M_{i,j}$ depends only on the coordinate pair (i,j) and the column parameter A_j, every cell is computable in $\mathcal{O}(1)$ arithmetic independently of all others, with no inter-row dependencies.

6.1 Thread Organization

The matrix $M \in Z^{n\times m}$ is laid out in row-major order as a one-dimensional array of $N = n \times m$ cells. Threads are organized in a two-dimensional grid: each thread is identified by a pair (t_x, t_y) within a block of dimensions $B_x \times B_y = 32 \times 8$, yielding 256 threads per block. The grid dimensions are

$$G_x = \left\lceil \frac{m}{B_x} \right\rceil, \qquad G_y = \left\lceil \frac{n}{B_y} \right\rceil, \tag{1}$$

so that the global thread indices

$$j = \text{blockIdx}_x \cdot B_x + t_x, \qquad i = \text{blockIdx}_y \cdot B_y + t_y \tag{2}$$

map bijectively onto the matrix coordinates (i,j), with out-of-bound threads discarded via a boundary guard. The total number of active threads equals exactly $n \times m$, one per cell.

6.2 Task 1 : Parallel Matrix Construction

The first parallelized task constructs both compressed representations of M — the log-magnitude array $\ell_{i,j} = \log M_{i,j}$ and the modular residue array $r_{i,j} = M_{i,j} \bmod 6$ — by exploiting the fact that both quantities admit closed-form expressions depending only on the coordinate pair (i, j) and the column base value A_j, with no dependency on any neighboring cell. Each thread independently recovers A_j by iterating over qualifying odd integers up to the j-th element, derives the column seed $A_j^{(1)}$ according to column parity, computes the log-magnitude via a numerically stable log-sum-exp formulation with a negligibility threshold on the correction term, and evaluates the modular residue through a closed-form period-2 arithmetic expression. Since all four steps are entirely local to thread (i, j), this phase requires no synchronization, no shared memory, and no inter-thread communication of any kind, with each thread performing $\mathcal{O}(j)$ work to recover A_j and $\mathcal{O}(1)$ arithmetic for all remaining computations.

6.3 Task 2 : Parallel Maximum Identification

The second parallelized task identifies the index pair (i^*, j^*) such that $\ell_{i^*,j^*} = \max_{i,j} \ell_{i,j}$. The core challenge is efficiently reducing a large array to a single extremal value across many parallel threads.

The flattened array $\{\ell_{i,j}\}$ of length $N = nm$ is partitioned into fixed-size blocks of threads. The reduction proceeds in two stages: within each block, threads collaboratively compute a local maximum by iteratively halving the number of active participants until a single (value, index) pair survives per block. The partial maxima produced by all blocks are then gathered and subjected to a second, structurally identical reduction pass, yielding the global maximum and its position in $\mathcal{O}(\log N)$ total steps. The corresponding coordinates (i^*, j^*) and the exact value ℓ_{i^*,j^*} are subsequently recovered on the host via a closed-form index inversion in $\mathcal{O}(1)$.

6.4 Task 3 : Parallel Index Extraction

The third parallelized task addresses a classical stream-compaction problem: collecting all cell coordinates (i, j) where the residue $r_{i,j} = 3$ into a contiguous output array, without prior knowledge of the output size and without write conflicts between threads. The key insight is to separate counting from writing across two passes. In the first pass, each thread group independently evaluates the predicate and counts its local qualifying elements, with no inter-group communication whatsoever. Between the two passes, the host computes an exclusive prefix sum over these local counts, which assigns every thread group a pre-determined, non-overlapping slice of the output array before a single result is written. In the second pass, each thread re-evaluates its predicate and writes directly to its pre-assigned position, so that every output address is fully resolved prior to any write, eliminating the need for atomic operations or any form of contention arbitration entirely.

The implementation exploits GPU parallelism to construct and analyze an odd-matrix of shape $n_{\text{rows}} \times n_{\text{cols}}$, where each column j is seeded by the $(j+1)$-th odd integer not divisible by 3, denoted A_j.

6.5 Scalability Analysis

For a square matrix of dimension $n \times n$, the thread and block organization scales as follows. The two-dimensional block size is fixed at $B_x \times B_y = 32 \times 8 = 256$ threads per block, and the grid dimensions grow linearly with n:

$$G_x = \left\lceil \frac{n}{32} \right\rceil, \qquad G_y = \left\lceil \frac{n}{8} \right\rceil, \tag{3}$$

so that the total number of launched threads is

$$T(n) = \left\lceil \frac{n}{32} \right\rceil \cdot \left\lceil \frac{n}{8} \right\rceil \cdot 256 \;\approx\; n^2 \quad \text{for large } n. \tag{4}$$

Table 4 reports the exact thread and block counts for representative matrix sizes.
The practical upper bound on n is governed by two hardware constraints. First, the maximum number of blocks per grid dimension on current architectures is $2^{31} - 1$, which is never the limiting factor here. Second, and more critically, the two output arrays — the log-magnitude array of n^2 double-precision values and the residue array of n^2 bytes — must reside simultaneously in device global memory, imposing the memory constraint

$$n^2 \cdot (8 + 1) \;\leq\; \text{VRAM}_{\text{available}}, \tag{5}$$

which gives, for example, $n \leq 33\,000$ on an 8 GB device and $n \leq 47\,000$ on a 16 GB device.

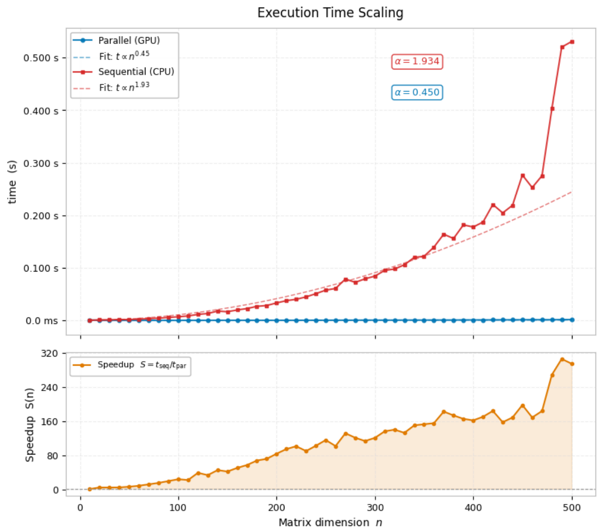

Figure 3: Excution time scalling

Matrix size $n \times n$	**Total cells**	**Grid** $(G_x \times G_y)$	**Blocks**	**Active threads**
64×64	4 096	2×8	16	4 096
128×128	16 384	4×16	64	16 384
256×256	65 536	8×32	256	65 536
500×500	250 000	16×63	1 008	258 048
1024×1024	1 048 576	32×128	4 096	1 048 576
2048×2048	4 194 304	64×256	16 384	4 194 304
4096×4096	16 777 216	128×512	65 536	16 777 216

Table 4: Thread and block scaling for representative square matrix sizes $n \times n$, with fixed block dimensions 32×8.

Figure 3 illustrates this constraint empirically. The sequential (CPU) curve grows much faster than linearly ($\alpha \approx 1.93$), and over its last four data points the execution time rises sharply for only modest increases in n — a clear sign of poor scalability, where resource limits (memory bandwidth, cache effects) begin to dominate over the underlying $\mathcal{O}(n^2)$ arithmetic cost. The parallel (GPU) curve, by contrast, stays almost flat ($\alpha \approx 0.45$) across the same range, confirming that distributing the workload across threads keeps execution time largely decoupled from n. This asymmetry is reflected directly in the speedup curve, which grows steadily and spikes precisely where the sequential implementation degrades — underscoring that parallelization here is not just faster, but fundamentally more scalable.

7 Conclusion

We have identified and documented a structured sub-series within the Odd Numbers Matrix of the Collatz Conjecture: the family of odd integers of the form $3(2n+1)$ is systematically excluded from the first column of the matrix (no odd integer maps directly to any member of this family), yet appears in later columns with a regular, wave-like pattern of column positions. This observation is non-trivial, admits a partial arithmetic proof, and raises concrete open questions about the precise characterisation of the wave pattern and its implications for convergence.

In the spirit of the broader mathematical programme of identifying structured sub-series within complex sequences, we hope that this note draws attention to a feature of the Collatz dynamics that merits further investigation. Whether or not this particular sub-series ultimately contributes to a resolution of the conjecture, its regularity is a reminder that even in the most intractable problems, structure is waiting to be found.

A Scalability Details

For completeness, the appendix provides a more detailed account of the scalability behaviour of the parallel program as the input size grows. Specifically, Table 5 reports the monitoring metrics recorded during execution for square matrices of increasing dimension $n \times n$ using PyCUDA. The columns track RAM usage, the number of threads that were actually active, the total number of threads launched by the GPU kernel, and the maximum value achieved by the computation.

Table 5: Monitoring metrics as n increases ($n \times n$ matrix)

n	RAM (%)	Threads active	Threads launched	Max value
50	1.8	2,500	3,584	$\approx 2^{103.65}$
100	1.8	10,000	13,312	$\approx 2^{204.65}$
150	1.8	22,500	24,320	$\approx 2^{305.23}$
200	1.8	40,000	44,800	$\approx 2^{405.65}$
250	1.8	62,500	65,536	$\approx 2^{505.97}$
300	1.8	90,000	97,280	$\approx 2^{606.23}$
350	1.8	122,500	123,904	$\approx 2^{706.45}$
400	1.8	160,000	166,400	$\approx 2^{806.65}$
450	1.8	202,500	218,880	$\approx 2^{906.81}$
500	1.8	250,000	258,048	$\approx 2^{1006.97}$
550	1.8	302,500	317,952	$\approx 2^{1107.10}$

A notable observation from Table 5 is that RAM consumption remains essentially constant at 1.8% across all tested sizes, confirming that the program's memory footprint scales efficiently with n. The number of active threads grows quadratically ($\propto n^2$), as expected for a matrix-based workload, while the maximum value grows approximately linearly in its binary exponent ($\approx 2n$ bits per unit increase in n).

Hardware limit. All experiments were conducted on the Google Colab GPU, a NVIDIA Tesla T4 clocked at 1.59 GHz, equipped with 8 GB of device memory and a shared cache of 6 MB. Under this hardware configuration, the practical upper bound we were able to reach is $n < 600$. Attempting to process a 600×600 matrix (or larger) caused the kernel to exceed the available GPU resources, resulting in a runtime failure.

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